

Shape 2: an ellipsoid with an offset centre of mass

Does moving the CoM off the centre of force, in any direction, give the chirality needed for orbits?

The body is the near-spherical ellipsoid of Shape 1, now with a small linear density gradient that moves the centre of mass (CoM) off the geometric centre H. Since the ellipsoid is centrosymmetric, H is also its centre of force. As before, the gradient is uniform, $\nabla \ln T = -G \hat{Z}$, with no trap. The response $(\mathbf{F}, \boldsymbol{\tau}) = -\mathcal{R} \cdot (\nabla \ln T, \mathbf{v}, \boldsymbol{\omega})$ is taken about the CoM.

Results in brief. (1) An offset along a principal axis gives symmetry C_{2v} , and an offset in a principal plane gives C_s . On a triaxial ellipsoid an offset in a generic direction leaves no symmetry at all (C_1), so the particle **is chiral**. (2) All four tensors follow exactly from those of Shape 1 by a rigid shift. In particular $\mathbf{T}_{th} = -[\mathbf{r}_c]_x \mathbf{f}_{th}$. (3) This chirality is invisible to the gas. $\mathbf{T}_{drag}^{-1} \mathbf{T}_{th}$ never has a real non-zero eigenvalue, whatever the offset direction. The reason: at levitation the net gas force is vertical and acts through H, so it has no moment about the vertical. (4) The particle hangs with its CoM below H, as a damped pendulum, and then **glides in a straight line** (offset in a mirror plane or generic) or hovers (offset along an axis). (5) There is **no orbit** for any offset: $R = \infty$ and there is no period.

1. Setup

```

ClearAll["Global`*"];
shapesDir = Quiet@Check[NotebookDirectory[], $Failed];
If[! StringQ[shapesDir], shapesDir = Environment["SHAPES_DIR"]];
Get[FileNameJoin[{shapesDir, "ShapeLevitation.wl"}]];
$verifications = {};
record[ref_, claim_, status_, detail_] :=
  (AppendTo[$verifications, {ref, claim, status, detail}];
   VR[ref, claim, status, detail]);
VR /: MakeBoxes[VR[ref_, claim_, status_, detail_, fmt_] :=
  ToBoxes[Panel[Column[{Row[{Style[status, Bold,
    If[status == "VERIFIED", Darker[Green], Red]], " ", Style[ref, Bold],
    ": ", claim}], If[detail == "", Nothing, detail]}]], fmt];
check[ref_String, claim_String, test_, detail_ : "" ] :=
  record[ref, claim, If[TrueQ[test], "VERIFIED", "DISCREPANCY"], detail];
zeroQ[expr_] := AllTrue[Flatten[{Simplify[expr]}], PossibleZeroQ];
sci[x_] := If[x == 0, "0", Module[{e = Floor[Log10[Abs[N[x]]]}],
  ToString[NumberForm[N[x] / 10^e, {3, 2}]] <> "e" <> ToString[e]]];
num[x_, d_ : 4] := If[x != 0 && (Abs[x] < 10^-3 || Abs[x] ≥ 10^5),
  sci[x], ToString[NumberForm[N[x], d]]];
vec[v_, d_ : 3] := "(" <> StringRiffle[num[#, d] & /@ v, ", " <> ")";
tf[m_] := TraditionalForm[MatrixForm[m]];
n = {x, y, z};
gas = gasModel[];
ell = 20*^-6;
rhoP = 1000.;

```

2. The shape and the mass distribution

The surface is the ellipsoid of Shape 1, with semi-axes $a_i = 1 + \varepsilon e_i$ (units of l), defaults $\varepsilon = 0.1$ and $e = (1, 0, -1)$. The density is $\rho_p(1 + \kappa \hat{d} \cdot \mathbf{r}/l)$, with unit direction $\hat{d}$ and strength κ . Because $\oint r_i r_j dV = \frac{V}{5} a_i^2 l^2 \delta_{ij}$ for an ellipsoid, the CoM sits at

$$\mathbf{r}_c = \frac{\kappa l}{5} \text{diag}(a_1^2, a_2^2, a_3^2) \hat{d} \quad (\text{exact}; \quad |\mathbf{r}_c| \approx 0.1 l = 2 \mu\text{m for } \kappa = 0.5)$$

```

a1 = 1 + eps e1; a2 = 1 + eps e2; a3 = 1 + eps e3;
Rellipsoid[{x_, y_, z_}] := 1 / Sqrt[x^2 / a1^2 + y^2 / a2^2 + z^2 / a3^2];
density[r_, dhat_, kappa_] := 1 + kappa dhat.r;
defaults = {eps → 1 / 10, e1 → 1, e2 → 0, e3 → -1}; kappa0 = 0.5;
Rnum[nn_List] := Rellipsoid[nn] /. defaults;
offsets = <|"along x (C2v)" → {1, 0, 0}, "in the x-z plane (Cs)" →
  Normalize[{1, 0, 1}], "generic (C1, chiral)" → Normalize[{1, 2, 3}]|>;
Row[{"R(n) = ", TraditionalForm[Rellipsoid[n]], ", rho/rho_p = ",
  TraditionalForm[density[{X, Y, Z}, {d1, d2, d3}, kappa]],
  ", defaults ", defaults, ", kappa = ", kappa0]}

```

$$R(n) = \frac{1}{\sqrt{\frac{x^2}{(e1 \text{ eps} + 1)^2} + \frac{y^2}{(e2 \text{ eps} + 1)^2} + \frac{z^2}{(e3 \text{ eps} + 1)^2}}}, \quad \text{rho/rho}_p = \text{kappa} (d1 X + d2 Y + d3 Z) + 1$$

, defaults {eps → $\frac{1}{10}$, e1 → 1, e2 → 0, e3 → -1}, kappa = 0.5

Three offset directions represent the three symmetry classes that an offset can produce on a triaxial ellipsoid:

offset direction $\hat{d}$	symmetry of body + mass	surviving elements
along a principal axis, e.g. $\hat{x}$	C_{2v}	the two – fold axis along $\hat{d}$ and two r
in a principal plane, e.g. $(\hat{x} + \hat{z}) / \sqrt{2}$	C_s	the mirror plane containing $\hat{d}$
generic, e.g. $(1, 2, 3) / \sqrt{14}$	C_1	none

(For a spheroid, $a_1 = a_2$, even a generic offset leaves the mirror plane that contains the symmetry axis and $\hat{d}$. So only a triaxial ellipsoid can be made chiral this way.) Next, build the three particles by numerical quadrature, and check the CoM formula and convexity.

```

pts = Association[KeyValueMap[
  #1 → makeParticle[Rnum, ell, rhoP, gas, Function[r, density[r, #2, kappa0]]] &,
  offsets]];
pt0 = makeParticle[Rnum, ell, rhoP, gas];
rcExact[dhat_] := kappa0 ell / 5 DiagonalMatrix[N[{a1, a2, a3}^2 /. defaults]].dhat;
Column[Join[
  {check["convex", "the surface (Shape 1) is convex", convexityMargin[Rnum] > 0]},
  KeyValueMap[check["CoM " <> #1, "CoM = (kappa l/5) diag(a_i^2).d",
    Norm[#2["com"] - rcExact[offsets[#1]]] < 10^-9 ell,
    "r_c/l = " <> vec[#2["com"] / ell, 4]] &, pts]]]

```

✓ **VERIFIED convex:** the surface (Shape 1) is convex

✓ **VERIFIED CoM along x (C2v):** CoM = $(\kappa l/5) \text{diag}(a_i^2).d$

$r_c/l = (0.121, -8.46e-20, -2.97e-17)$

✓ **VERIFIED** CoM in the x-z plane (Cs): $\text{CoM} = (\kappa l/5) \text{diag}(a_i^2).d$

$r_c/l = (0.08556, -2.21e-18, 0.05728)$

✓ **VERIFIED** CoM generic (C1, chiral): $\text{CoM} = (\kappa l/5) \text{diag}(a_i^2).d$

$r_c/l = (0.03234, 0.05345, 0.06494)$

3. Diagram

Each panel shows the body coloured by $R - 1$ (red is long), the geometric centre H (black) and the CoM (magenta). The magenta arrow is r_c drawn 5 times longer than scale. In the chiral case the arrow leans out of every principal plane.

Row[KeyValueMap[

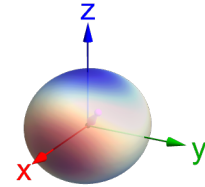
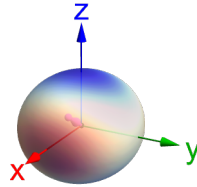
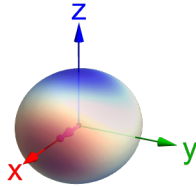
Show[shapeDiagram[Rnum, 5 #2["com"] / ell, PlotLabel -> #1, ImageSize -> 205],

Graphics3D[{Magenta, Arrow[Tube[{{0, 0, 0}, 5 #2["com"] / ell}, 0.03]]}] &, pts]]

along x (C2v)

in the x-z plane (Cs)

generic (C1, chiral)



4. The four tensors

The gas sees only the surface. Every stress therefore acts exactly as in Shape 1, about H, where

$T_{th}^H = B^H = 0$. Moving the reference point from H to the CoM changes the lever arm, $r_H = -r_c$, and the velocity of H, $v_H = v + \omega \times r_H$. The result is exact:

$$f_{th} = f_{th}^H, \quad f_{drag} = f_{drag}^H, \quad B = f_{drag} [r_c]_{\times}, \quad B^T = -[r_c]_{\times} f_{drag}$$

$$T_{th} = -[r_c]_{\times} f_{th} \approx -\gamma_0 [r_c]_{\times}, \quad T_{drag} = T_{drag}^H + [r_c]_{\times}^T f_{drag} [r_c]_{\times}$$

Here f_{th}^H , f_{drag}^H and T_{drag}^H are the diagonal first-order forms of Shape 1. At $\alpha = 1$ they are

$\gamma_0 \text{diag}[1 + \frac{2\varepsilon}{5} (2E - e_i)]$, $\xi_0 \text{diag}[1 + \frac{2\varepsilon}{5} (3E - 4e_i + 8(3e_i - E)/(8 + \pi))]$ and $c_0 \text{diag}[1 + \frac{4\varepsilon}{5} (2E - e_i)]$, with $E = \sum e_i$. The thermophoretic torque is **antisymmetric** to leading order: it is the moment about

the CoM of a force $\gamma_0 G \mathbf{u}$ applied at H. The check below compares the full numerical quadrature about the CoM with these shift formulas.

```
shiftRule[p0_, rc_] :=
  Module[{b = p0["blocks"], X = hat[rc]}, <|"f_th" → b["f_th"], "f_drag" → b["f_drag"],
    "B" → b["f_drag"].X, "Bt" → -X.b["f_drag"], "T_th" → -X.b["f_th"],
    "T_drag" → b["T_drag"] + Transpose[X].b["f_drag"].X|>];
Column[KeyValueMap[
  Module[{sr = shiftRule[pt0, #2["com"]], b = #2["blocks"]}, check["shift "<> #1,
    "quadrature tensors about the CoM equal the shift of the Shape 1 tensors",
    Max[Norm[b["f_th"] - sr["f_th"]] / pt0["gamma0"],
      Norm[b["f_drag"] - sr["f_drag"]] / pt0["xi0"],
      Norm[b["B"] - sr["B"]] / (pt0["xi0"] ell), Norm[b["Bt"] - sr["Bt"]] /
        (pt0["xi0"] ell), Norm[b["T_th"] - sr["T_th"]] / (pt0["gamma0"] ell),
      Norm[b["T_drag"] - sr["T_drag"]] / pt0["c0"]] < 10^-9]] &, pts]]
```

✓ **VERIFIED shift along x (C2v):** quadrature tensors about the CoM equal the shift of the Shape 1 tensors

✓ **VERIFIED shift in the x-z plane (Cs):** quadrature tensors about the CoM equal the shift of the Shape 1 tensors

✓ **VERIFIED shift generic (C1, chiral):** quadrature tensors about the CoM equal the shift of the Shape 1 tensors

The same result follows symbolically. The next cell runs the perturbation engine with the density written as $1 + \varepsilon(k_1 x + k_2 y + k_3 z)$, so that the shape and the offset are both of order ε . It confirms $\mathbf{T}_{th} = -[r_c]_x \mathbf{f}_{th}$ through second order, including the shape $\times$ offset cross terms.

```
shS =
  symbolicShape[1 + eps (e1 x^2 + e2 y^2 + e3 z^2), n, eps, 2, 1 + eps (k1 x + k2 y + k3 z)];
Module[{b = shS["blocks"], rc = shS["com"], trunc2},
  trunc2[m_] := Map[Normal[Series[#, {eps, 0, 2}]] &, m, {2}];
Column[{Row[{"r_c/l = ", TraditionalForm[Simplify[rc]]}], Row[{"(T_th)_12/(gamma0
  l) = ", TraditionalForm[Simplify[b["T_th"][[1, 2]]]}], check[
  "symbolic", "T_th = -[r_c]x f_th through O(eps^2), with r_c = (eps/5)(k_i
    (1 + 2 eps e_i))", zeroQ[trunc2[b["T_th"] + hat[rc].b["f_th"]]] &&
    zeroQ[rc - eps/5 Table[{k1, k2, k3}[[i]] (1 + 2 eps {e1, e2, e3}[[i]), {i, 3}]]]]]
  r_c/l = {1/5 eps k1 (2 e1 eps + 1), 1/5 eps k2 (2 e2 eps + 1), 1/5 eps k3 (2 e3 eps + 1)}
  (T_th)_12/(gamma0 l) = 1/25 eps k3 (4 e1 eps + 2 e2 eps + 14 e3 eps + 5)
```

✓ **VERIFIED symbolic:** $T_{th} = -[r_c]_x f_{th}$ through $O(\varepsilon^2)$, with $r_c = (\varepsilon/5)(k_i (1 + 2 \varepsilon e_i))$

The default particles give these numbers (SI; the offset breaks no symmetry of $\mathbf{f}_{\text{th}}$ or $\mathbf{f}_{\text{drag}}$):

```
Grid[Prepend[
  KeyValueMap[{#1, tf[Round[#2["blocks"] ["T_th"] / (#2["gamma0"] ell), 10.^-4]],
    tf[Round[#2["blocks"] ["T_drag"] / #2["c0"], 10.^-4]]} &, pts],
  {"offset", "T_th/(gamma0 l)", "T_drag/c0"}], Frame -> All, Alignment -> Center]
```

offset	T_th/(gamma0 l)	T_drag/c0
along x (C2v)	$\begin{pmatrix} 0. & 0. & 0. \\ 0. & 0. & 0.1259 \\ 0. & -0.1204 & 0. \end{pmatrix}$	$\begin{pmatrix} 0.93 & 0. & 0. \\ 0. & 1.0873 & 0. \\ 0. & 0. & 1.1287 \end{pmatrix}$
in the x-z plane (Cs)	$\begin{pmatrix} 0. & 0.057 & 0. \\ -0.055 & 0. & 0.089 \\ 0. & -0.0852 & 0. \end{pmatrix}$	$\begin{pmatrix} 0.9391 & 0. & -0.0136 \\ 0. & 1.0738 & 0. \\ -0.0136 & 0. & 1.1084 \end{pmatrix}$
generic (C1, chiral)	$\begin{pmatrix} 0. & 0.0646 & -0.0556 \\ -0.0624 & 0. & 0.0336 \\ 0.0513 & -0.0322 & 0. \end{pmatrix}$	$\begin{pmatrix} 0.9502 & -0.0052 & -0.0058 \\ -0.0052 & 1.0575 & -0.009 \\ -0.0058 & -0.009 & 1.0985 \end{pmatrix}$

5. Why a mass offset cannot make the particle spin

A steady spin $\boldsymbol{\omega} = \Omega \mathbf{u}$ about the vertical needs a real eigenpair $\mathbf{T}_{\text{drag}}^{-1} \mathbf{T}_{\text{th}} \mathbf{u} = \lambda \mathbf{u}$ with $\lambda \neq 0$ (then $\Omega = \lambda G$). Insert $\mathbf{T}_{\text{th}} = -[\mathbf{r}_c]_{\times} \mathbf{f}_{\text{th}}$ and take the dot product with $\mathbf{f}_{\text{th}} \mathbf{u}$:

$$\lambda \mathbf{u}^T \mathbf{f}_{\text{th}} \mathbf{T}_{\text{drag}} \mathbf{u} = -(\mathbf{f}_{\text{th}} \mathbf{u}) \cdot (\mathbf{r}_c \times \mathbf{f}_{\text{th}} \mathbf{u}) = 0 \Rightarrow \lambda = 0$$

The conclusion holds whenever the symmetric part of $\mathbf{f}_{\text{th}} \mathbf{T}_{\text{drag}}$ is positive-definite, which is always true for a near-sphere ($\approx \gamma_0 c_0 \mathbf{I}$). The only real eigenvalue is therefore $\lambda = 0$, with $\mathbf{f}_{\text{th}} \mathbf{u} \parallel \mathbf{r}_c$: the hanging orientation. The other two eigenvalues form a complex pair $\approx \pm i \gamma_0 |\mathbf{r}_c| / c_0$, a pendulum oscillation. Physically, every gas force acts through H. At levitation the total gas force balances gravity and is vertical, and a vertical force through H has no moment about the vertical axis through the CoM. So there is never a spin-driving torque, whichever way the offset points.

```
Column[KeyValueMap[
  Module[{ev = Eigenvalues[Inverse[#2["blocks"] ["T_drag"]].#2["blocks"] ["T_th"]],
    mu = #2["gamma0"] Norm[#2["com"]] / #2["c0"]}, check["spectrum " <> #1,
    "eigenvalues of T_drag^-1 T_th are 0 and a complex pair ~ +-i gamma0
      |r_c|/c0", Count[ev, e_ /; Abs[Im[e]] < 10^-9 Abs[mu]] == 1 &&
      Min[Abs[ev]] < 10^-9 mu && Abs[Max[Abs[Im[ev]]] / mu - 1] < 0.1 &&
      Min[Eigenvalues[(#2["blocks"] ["f_th"].#2["blocks"] ["T_drag"] +
        Transpose[#2["blocks"] ["f_th"].#2["blocks"] ["T_drag"]]) / 2]] > 0,
    "eigenvalues/(gamma0 |r_c|/c0) = " <> StringRiffle[
      ToString[NumberForm[Chop[# / mu, 10^-10], 4]] & /@ ev, ", "]] &, pts]]]
```

✓ **VERIFIED spectrum along x (C2v):** eigenvalues of $T_{\text{drag}}^{-1} T_{\text{th}}$ are 0 and a complex pair $\sim \pm i \gamma_0 |r_c|/c_0$

eigenvalues/ $(\gamma_0 |r_c|/c_0) = 0. + 0.9186 i, 0. - 0.9186 i, 0$

✓ **VERIFIED spectrum in the x-z plane (Cs):** eigenvalues of $T_{\text{drag}}^{-1} T_{\text{th}}$ are 0 and a complex pair $\sim \pm i \gamma_0 |r_c|/c_0$

eigenvalues/ $(\gamma_0 |r_c|/c_0) = 0. + 0.9397 i, 0. - 0.9397 i, 0$

✓ **VERIFIED spectrum generic (C1, chiral):** eigenvalues of $T_{\text{drag}}^{-1} T_{\text{th}}$ are 0 and a complex pair $\sim \pm i \gamma_0 |r_c|/c_0$

eigenvalues/ $(\gamma_0 |r_c|/c_0) =$
 $-4.762 \times 10^+ 0.9666 i, -6$
 $-4.762 \times 10^- 0.9666 i, 0$

6. Steady states: hang, then glide or hover

With no spin available, the steady states are non-rotating. The solver **steadyMotions** imposes exact body-frame balance of force and torque, including the drag torque of the glide velocity about the CoM ($\mathbf{B}^T \mathbf{v}$), together with $\mathbf{u} \cdot \mathbf{v} = 0$. It then linearises the full rigid-body equations about each solution. For each offset it finds two states: CoM below H (stable) and CoM above H (unstable).

```
steady = Association[KeyValueMap[#1 → steadyMotions[#2] &, pts]];
hang = Association[KeyValueMap[#1 → First@Select[#2, #["stable"] &] &, steady]];
Grid[Prepend[KeyValueMap[{#1, vec[Chop[#2["u"], 10^-9]], num[#2["Omega"]],
  num[1000 Chop[#2["vperp"], 10^-9]], If[#2["vperp"] > 10^-9,
    vec[Normalize[#2["v_body"] - (#2["v_body"].#2["u"]) #2["u"]]], "-"],
    num[Max[Abs[Im[#2["rates"]]]] / (2 Pi)]] &, hang],
  {"offset", "u = lab Z in body axes", "Omega (rad/s)", "glide (mm/s)",
    "glide direction (body)", "pendulum f (Hz)"},
  Frame → All, Alignment → Center]
```

offset	u = lab Z in body axes	Omega (rad/s)	glide (mm/s)	glide direction (body)	pendulum f (Hz)
along x (C2v)	(-1., 0., 0.)	0.	0.	-	60.34
in the x-z plane (Cs)	(-0.831, 0., -0.556)	0.	1.406	(0.556, -4.11e-17, -0.831)	53.83
generic (C1, chiral)	(-0.359, -0.593, -0.721)	0.	1.113	(0.691, 0.351, -0.632)	51.91

The hanging vertical is exactly $\mathbf{u} = -\hat{\mathbf{r}}_c$, with the CoM straight below H. The reason: in steady state the total gas force equals $Mg \mathbf{u}$ and acts through H, so its moment about the CoM vanishes only if H lies on

the vertical through the CoM. (The zero eigenvector of $\mathbf{T}_{th}$ alone, $\mathbf{f}_{th} \mathbf{u} \parallel \mathbf{r}_c$, differs slightly. The drag torque of the glide, $\mathbf{B}^T \mathbf{v}$, makes up the difference.) For the axial offset $\mathbf{u}$ is a principal axis and the particle hovers. For the in-plane and generic offsets it glides, at the speed Shape 1 predicts for that orientation, $\mathbf{v}_\perp \approx (Mg/\xi_0)[(\mathbf{f}_{th}/\gamma_0 - \mathbf{I})\mathbf{u}]_\perp$. In the mirror-symmetric (C_s) case the glide stays in the mirror plane. In the chiral (C_1) case it points in a generic direction. It is still a straight line.

```
Column[Join[KeyValueMap[Module[
  {u = #2["u"], rc = pts[#1]["com"], p = pts[#1], lo}, lo = p["M"] $gEarth / p["xi0"]
  Module[{w = (p["blocks"]["f_th"] / p["gamma0"] - IdentityMatrix[3]).u},
    Norm[w - (w.u) u]]];
  check["hang " <> #1, "stable state: Omega = 0, CoM exactly below H, glide
    speed matches the first-order formula", #2["Omega"] == 0 && u.rc < 0 &&
    VectorAngle[u, -rc] < 10^-6 && Abs[#2["vperp"] - lo] < 0.1 lo + 10^-9,
    "angle between u and -r_c: " <> num[VectorAngle[u, -rc] / Degree] <>
    " deg; glide " <> num[1000 #2["vperp"]] <> " mm/s, formula " <>
    num[1000 lo] <> " mm/s"]] &, hang], {check["mirror",
  "Cs offset: the glide lies in the mirror plane (y-component zero)",
  Abs[hang["in the x-z plane (Cs)"]["v_body"][[2]]] < 10^-12, check["unstable",
  "each offset also has an inverted state (CoM above H), which is unstable",
  AllTrue[Values[steady], Count[#, s_ /; ! s["stable"]] >= 1 &]]}]}
```

✓ **VERIFIED hang along x (C2v):** stable state: $\Omega = 0$, CoM exactly below H, glide speed matches the first-order formula

angle between u and $-\mathbf{r}_c$: 2.03e-14 deg; glide 2.39e-16 mm/s, formula 2.54e-16 mm/s

✓ **VERIFIED hang in the x-z plane (Cs):** stable state: $\Omega = 0$, CoM exactly below H, glide speed matches the first-order formula

angle between u and $-\mathbf{r}_c$: 1.14e-13 deg; glide 1.406 mm/s, formula 1.423 mm/s

✓ **VERIFIED hang generic (C1, chiral):** stable state: $\Omega = 0$, CoM exactly below H, glide speed matches the first-order formula

angle between u and $-\mathbf{r}_c$: 1.57e-13 deg; glide 1.113 mm/s, formula 1.123 mm/s

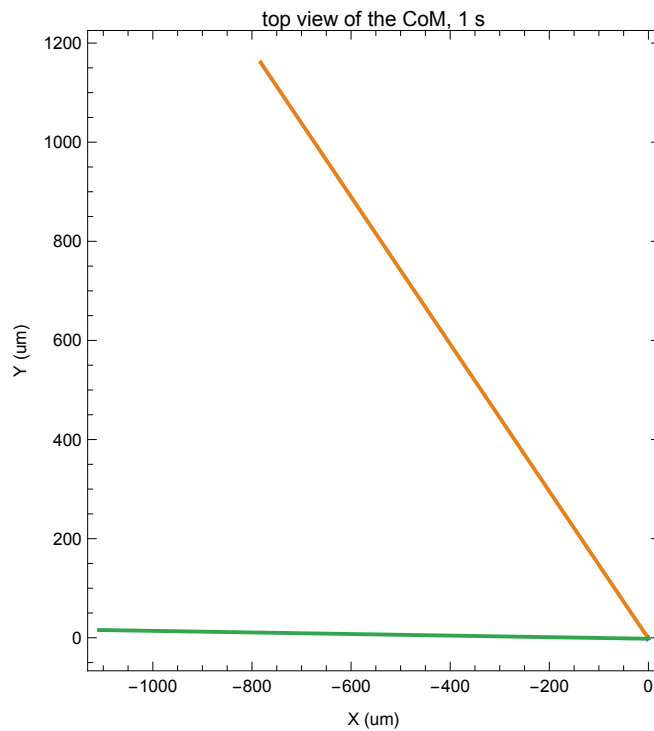
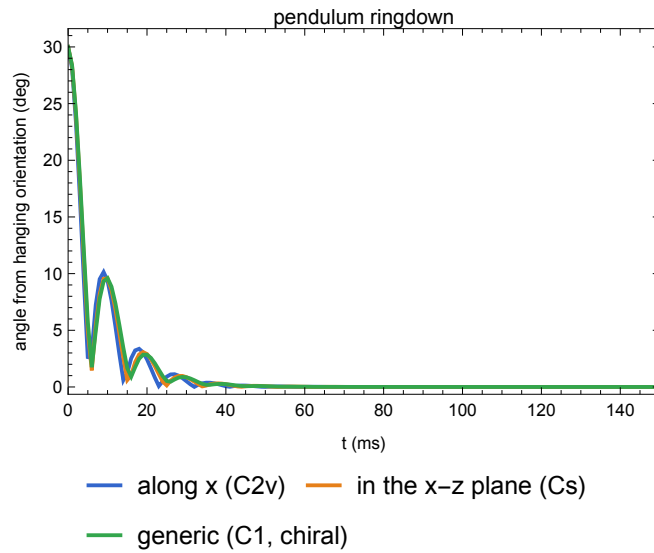
✓ **VERIFIED mirror:** Cs offset: the glide lies in the mirror plane (y-component zero)

✓ **VERIFIED unstable:** each offset also has an inverted state (CoM above H), which is unstable

7. Simulated trajectories

Each particle starts at rest, with its vertical tilted 30° away from the hanging orientation, at the hanging state's levitation gradient. The full nonlinear equations show a damped pendulum swing, with no net spin, followed by steady straight-line motion. That motion is a hover for the axial offset and a glide for the other two.

```
runs = Association[KeyValueMap[Module[{u = #2["u"], q0, tiltAxis},
  tiltAxis = Normalize[Cross[u, If[Abs[u[[2]]] < 0.9, {0, 1, 0}, {1, 0, 0}]]];
  q0 = qFromTo[RotationMatrix[30 Degree, tiltAxis].u, {0, 0, 1}];
  #1 → simulate[pts[#1], #2["G"], q0, {0, 0, 0}, 1.] &, hang]];
uLab[st_] := Transpose[quatR[st[[7 ;; 10]]].{0, 0, 1}];
Module[{ts = Subdivide[0., 1., 1000], cols =
  {RGBColor[0.2, 0.4, 0.8], RGBColor[0.9, 0.5, 0.1], RGBColor[0.2, 0.65, 0.3]}},
Column[{Row[{ListLinePlot[KeyValueMap[Function[{key, hs}, Transpose[
  {1000 ts, Table[VectorAngle[uLab[runs[key]]["state"][t]], hs["u"]] / Degree,
  {t, ts}]]]], hang], PlotRange → {{0, 150}, All}, Frame → True,
FrameLabel → {"t (ms)", "angle from hanging orientation (deg)"},
PlotLegends → Placed[Keys[hang], Below], PlotStyle → cols, ImageSize → 340,
PlotLabel → "pendulum ringdown"], ListLinePlot[KeyValueMap[
  Function[{key, hs}, Table[10^6 runs[key]]["state"][t][[1 ;; 2]], {t, ts}]],
  hang], AspectRatio → Automatic, Frame → True,
FrameLabel → {"X (um)", "Y (um)"}, PlotStyle → cols, ImageSize → 340,
PlotLabel → "top view of the CoM, 1 s", PlotRange → All]]],
Sequence @@ KeyValueMap[Module[{st = runs[#1]]["state"][1.], vpred},
  vpred = quatR[st[[7 ;; 10]]].#2["v_body"];
  check["run " <> #1, "after 1 s: no spin, orientation =
    hanging state, velocity = steady prediction",
  Norm[st[[11 ;; 13]]] < 10^-6 && VectorAngle[uLab[st], #2["u"]] < 10^-5 &&
  Norm[st[[4 ;; 6]] - vpred] < 10^-6 Max[Norm[vpred], 10^-6],
  "|omega| = " <> sci[Norm[st[[11 ;; 13]]]] <> " rad/s, horizontal speed " <>
  num[1000 Norm[st[[4 ;; 5]]]] <> " mm/s"] &, hang]]]
```



✓ **VERIFIED** run along x (C2v): after 1 s: no spin, orientation = hanging state, velocity = steady prediction

$|\omega| = 2.05\text{e-}14$ rad/s, horizontal speed $2.48\text{e-}15$ mm/s

✓ **VERIFIED** run in the x-z plane (Cs): after 1 s: no spin, orientation = hanging state, velocity = steady prediction

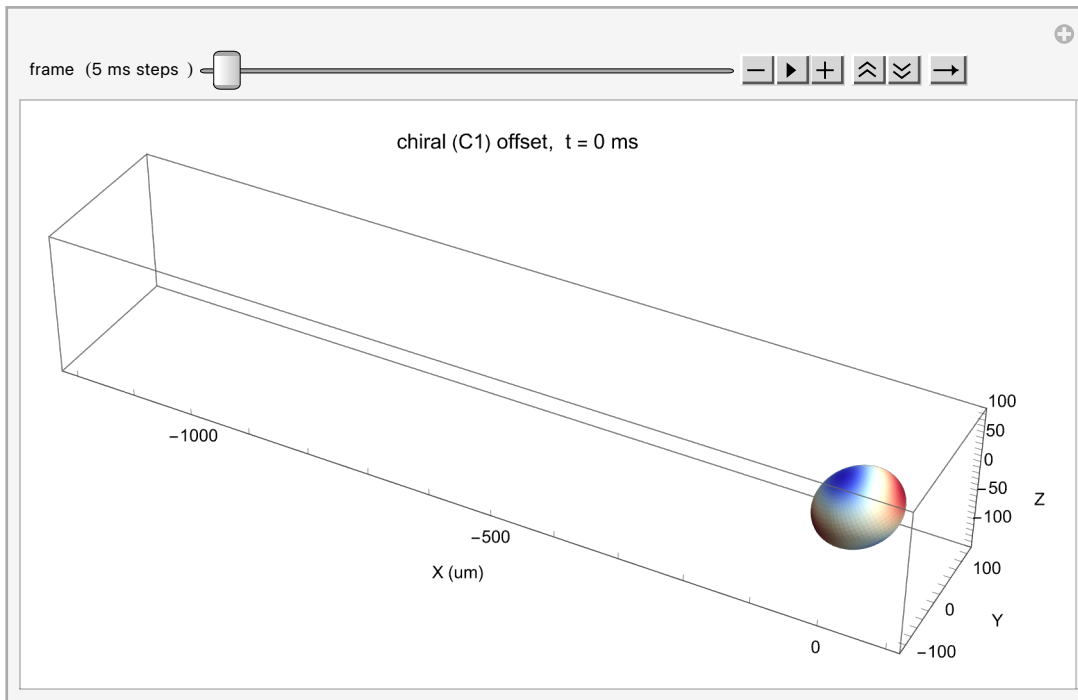
$|\omega| = 1.75\text{e-}12$ rad/s, horizontal speed 1.406 mm/s

✓ **VERIFIED** run generic (C1, chiral): after 1 s: no spin, orientation = hanging state, velocity = steady prediction

$|\omega| = 8.92e-13$ rad/s, horizontal speed 1.113 mm/s

The pendulum frequency is set by the stiffness $Mg |r_c|$ and the moment of inertia, as for the graded sphere. The ringdown takes a few tens of milliseconds. Afterwards the chiral particle moves exactly like an achiral one. Its orientation is locked, it has no spin, and its horizontal velocity is constant.

```
With[{mesh = bodyMesh[Rnum],
  states = Table[runs["generic (C1, chiral)"]["state"][t], {t, 0., 1., 0.005}],
  rng = 10^6 runs["generic (C1, chiral)"]["state"][1.][[1 ;; 3]],
  Manipulate[Module[{st = states[[k]], x}, x = 10^6 st[[1 ;; 3]],
    Graphics3D[{drawBody[mesh, x, quatR[st[[7 ;; 10]]], 60],
      If[k > 1, {Gray, Thick, Line[10^6 states[[1 ;; k, 1 ;; 3]]}, Nothing]],
      PlotRange → Transpose[{Min[#, 0] - 120 & /@ rng, Max[#, 0] + 120 & /@ rng}],
      BoxRatios → Automatic, Lighting → "Neutral", Axes → True, AxesLabel → {"X (um)",
        "Y", "Z"}, PlotLabel → Row[{"chiral (C1) offset, t = ", 5 (k - 1), " ms"}],
      ImageSize → 520]], {{k, 1, "frame (5 ms steps)"}, 1, 201, 1,
    ControlType → Animator, AnimationRunning → False}, SaveDefinitions → True]]
```



8. Orbital radius and period

Not applicable, for every offset direction. The steady states have $\Omega = 0$ exactly, so the orbital radius is infinite and there is no orbital period. The table below lists what the particle does instead.

```
Grid[
  Prepend[KeyValueMap[{#1, "0", "infinite", "none", If[#2["vperp"] < 10^-9, "hover",
    "straight glide at " <> num[1000 #2["vperp"]] <> " mm/s"]}] &, hang],
  {"offset", "spin Omega (rad/s)", "orbital radius", "period", "motion"}],
  Frame → All, Alignment → Left]
```

offset	spin Omega (rad/s)	orbital radius	period	motion
along x (C2v)	0	infinite	none	hover
in the x-z plane (Cs)	0	infinite	none	straight glide at 1.406 mm/s
generic (C1, chiral)	0	infinite	none	straight glide at 1.113 mm/s

9. Answer: can an offset direction give chirality?

- **Geometrically, yes.** On a triaxial ellipsoid, any offset that lies neither on a principal axis nor in a principal plane leaves no symmetry element. The particle (surface plus mass) is then chiral, C_1 .
- **Dynamically, no.** The gas sees only the surface, which is centrosymmetric, so the gas wrench about H has no polar-axial coupling. About the CoM the thermophoretic torque is the moment of a force through H, $\mathbf{T}_{th} = -[\mathbf{r}_c]_x \mathbf{f}_{th}$. Such a torque can only align the particle, never spin it: $\mathbf{T}_{drag}^{-1} \mathbf{T}_{th}$ has eigenvalues $\{0, \pm i\mu\}$ for every offset.
- **What is needed instead.** The chirality has to be in the *surface*, so that the gas forces themselves form a chiral pattern with a moment about the vertical. Shape 3 builds such a surface from spherical harmonics.

10. Summary

```
Column[{Row[{"checks: ", Length[$verifications], " ("},
  Count[$verifications[[All, 3]], "VERIFIED"], " verified)"}], Grid[Prepend[
  $verifications[[All, 1 ;; 3]], Style[#, Bold] & /@ {"item", "statement", "result"}],
  Frame → All, Alignment → Left, Background → {None, {LightGray}}}]
```

checks: 19 (19 verified)

item	statement	result
convex	the surface (Shape 1) is convex	VERIFIED
CoM along x (C2v)	$\text{CoM} = (\kappa l/5) \text{diag}(a_i^2).d$	VERIFIED
CoM in the x-z plane (Cs)	$\text{CoM} = (\kappa l/5) \text{diag}(a_i^2).d$	VERIFIED
CoM generic (C1, chiral)	$\text{CoM} = (\kappa l/5) \text{diag}(a_i^2).d$	VERIFIED
shift along x (C2v)	quadrature tensors about the CoM equal the shift of the Shape 1 tensors	VERIFIED
shift in the x-z plane (Cs)	quadrature tensors about the CoM equal the shift of the Shape 1 tensors	VERIFIED
shift generic (C1, chiral)	quadrature tensors about the CoM equal the shift of the Shape 1 tensors	VERIFIED
symbolic	$T_{\text{th}} = -[r_c]x f_{\text{th}}$ through $O(\epsilon^2)$, with $r_c = (\epsilon/5)(k_i(1 + 2\epsilon e_i))$	VERIFIED
spectrum along x (C2v)	eigenvalues of $T_{\text{drag}}^{-1} T_{\text{th}}$ are 0 and a complex pair $\sim \pm i \gamma_0 r_c /c_0$	VERIFIED
spectrum in the x-z plane (Cs)	eigenvalues of $T_{\text{drag}}^{-1} T_{\text{th}}$ are 0 and a complex pair $\sim \pm i \gamma_0 r_c /c_0$	VERIFIED
spectrum generic (C1, chiral)	eigenvalues of $T_{\text{drag}}^{-1} T_{\text{th}}$ are 0 and a complex pair $\sim \pm i \gamma_0 r_c /c_0$	VERIFIED
hang along x (C2v)	stable state: $\Omega = 0$, CoM exactly below H, glide speed matches the first-order formula	VERIFIED
hang in the x-z plane (Cs)	stable state: $\Omega = 0$, CoM exactly below H, glide speed matches the first-order formula	VERIFIED
hang generic (C1, chiral)	stable state: $\Omega = 0$, CoM exactly below H, glide speed matches the first-order formula	VERIFIED
mirror	Cs offset: the glide lies in the mirror plane (y-component zero)	VERIFIED
unstable	each offset also has an inverted state (CoM above H), which is unstable	VERIFIED
run along x (C2v)	after 1 s: no spin, orientation = hanging state, velocity = steady prediction	VERIFIED
run in the x-z plane (Cs)	after 1 s: no spin, orientation = hanging state, velocity = steady prediction	VERIFIED
run generic (C1, chiral)	after 1 s: no spin, orientation = hanging state, velocity = steady prediction	VERIFIED

- **Shape. Rellipsoid** of Shape 1 with density $1 + \kappa \text{dhat}.r$. Defaults: $\epsilon = 0.1$, $e = (1, 0, -1)$, $\kappa = 0.5$ ($|r_c| \approx 2 \mu\text{m}$).

- **Tensors.** $\mathbf{f}_{\text{th}}$ and $\mathbf{f}_{\text{drag}}$ are as in Shape 1. The others are $\mathbf{T}_{\text{th}} = -[\mathbf{r}_c]_{\times} \mathbf{f}_{\text{th}}$, $\mathbf{B} = \mathbf{f}_{\text{drag}} [\mathbf{r}_c]_{\times}$ and $\mathbf{T}_{\text{drag}} = \mathbf{T}_{\text{drag}}^H + [\mathbf{r}_c]_{\times}^T \mathbf{f}_{\text{drag}} [\mathbf{r}_c]_{\times}$, all exact.
- **Motion.** The particle swings as a damped pendulum into the CoM-below-H orientation. It then hovers (axial offset) or glides in a straight line (in-plane or generic offset).
- **Orbit.** None for any offset: a chiral mass distribution is not enough, and the chirality must be in the surface.